

Advanced (Habanero)

- Q1.** Which of the following is a condition for a function to have an inverse?
- (A) The function must be one-to-one
 - (B) The function must be a quadratic
 - (C) The function must be a polynomial
 - (D) The function must be periodic
 - (E) The function must be a constant
- Q2.** Given $f(x) = x^3 + 2x$, how can you find its inverse function $f^{-1}(x)$?
- (A) By solving $y = x^3 + 2x$ for x
 - (B) By graphing $y = x^3 + 2x$ and reflecting it across $y = x$
 - (C) By using the inverse function formula
 - (D) By differentiating $f(x)$
 - (E) By integrating $f(x)$
- Q3.** What is the purpose of verifying inverses using composition?
- (A) To show that the inverse is a function
 - (B) To show that the inverse is one-to-one
 - (C) To confirm that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$
 - (D) To find the domain of the inverse
 - (E) To find the range of the inverse
- Q4.** If $f(x) = 2x - 1$, what is $f^{-1}(x)$?
- (A) $\frac{x+1}{2}$
 - (B) $\frac{x-1}{2}$
 - (C) $2x + 1$
 - (D) $\frac{x}{2} - 1$
 - (E) $\frac{x+2}{2}$
- Q5.** Which of the following functions has an inverse?
- (A) $f(x) = x^2$
 - (B) $f(x) = |x|$
 - (C) $f(x) = x^3$
 - (D) $f(x) = \sin(x)$
 - (E) $f(x) = x^4 + 2x^2 + 1$
- Q6.** If $f(x) = 3x - 2$ and $g(x) = x + 2\frac{1}{3}$, what is $f(g(x))$?
- (A) x
 - (B) $x + 2$
 - (C) $x - 2$
 - (D) $2x + 1$
 - (E) $2x - 1$
- Q7.** What is the graphical representation of the inverse of $y = f(x)$?
- (A) A reflection across $y = x$
 - (B) A reflection across $y = -x$
 - (C) A rotation by 90°
 - (D) A translation by 2 units
 - (E) A scaling by a factor of 2
- Q8.** If $f(x)$ represents the amount of money in a bank account after x years, what does $f^{-1}(x)$ represent?
- (A) The amount of money in the account at the beginning of the year
 - (B) The number of years until the account reaches a certain amount
 - (C) The interest rate of the account
 - (D) The total amount of money deposited
 - (E) The total amount of money withdrawn

Open Ended Questions (FRQ)

- Q9.** Find the inverse of the function $f(x) = 2x + 1\frac{1}{x-1}$. Verify your answer using composition.
- Q10.** A water tank can be filled at a rate of $f(x) = 2x$ gallons per minute, where x is the number of minutes. Find the inverse function $f^{-1}(x)$ and interpret its meaning in this context.

Chef's Tips

- Remind students that the domain and range of the inverse are swapped compared to the original function
- Encourage students to check their inverses by composing them with the original function
- Emphasize the importance of understanding the concept of one-to-one functions in relation to inverses